

Report Labwork 3

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1 Load image

```
img_src = "/content/drive/MyDrive/DATA/HPC/image.jpg"  
  
img=plt.imread(img_src, format=None)  
  
plt.imshow(img)  
  
plt.show()
```

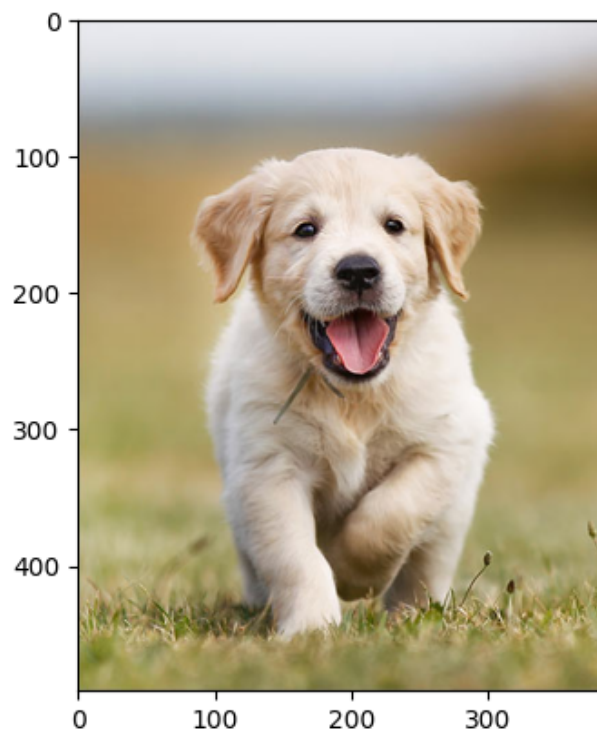


Figure 1: Load 320x320 image

2 Flatten image

Shape of flattened image: (190404, 3)

3 Grayscale

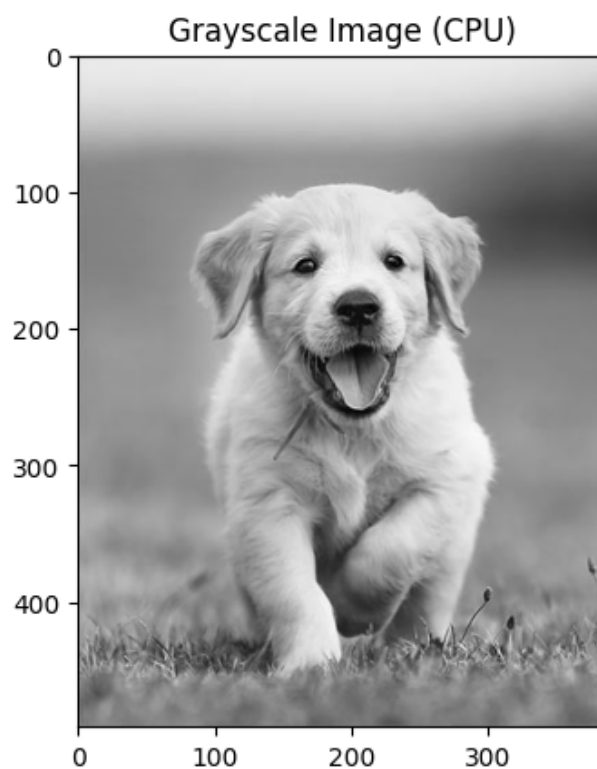
```
@cuda.jit
```

```
def grayscale_gpu(src, dst):
```

```
    tidx = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
```

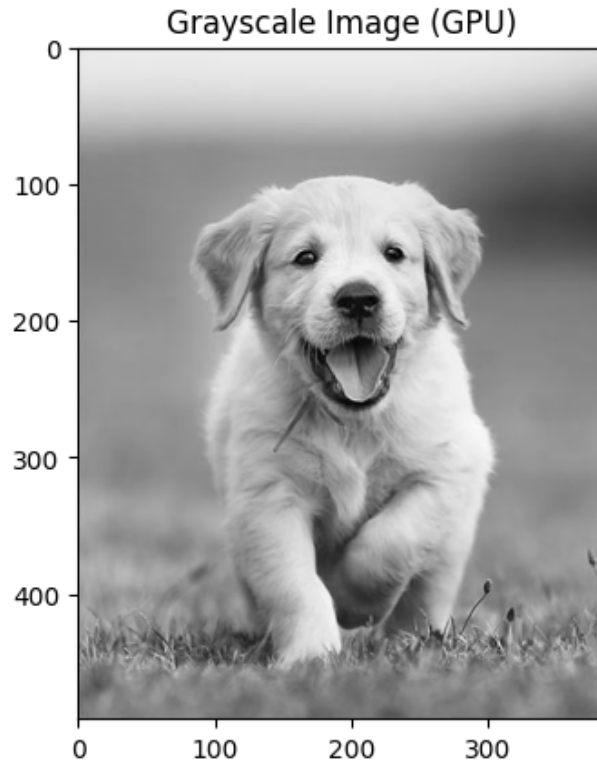
```
    g = np.uint8((src[tidx, 0] + src[tidx, 1] + src[tidx, 2]) / 3)
```

```
    dst[tidx, 0] = dst[tidx, 1] = dst[tidx, 2] = g
```



And grayscale for cpu:

```
def grayscale_cpu(image):  
    gray_image = np.zeros_like(image)  
  
    for i in range(image.shape[0]):  
        for j in range(image.shape[1]):  
            avg = np.mean(image[i, j])  
            gray_image[i, j] = [avg, avg, avg]  
  
    return gray_image
```



3.1 Result

CPU	GPU
1.3464 second	0.0004 second

Table 1: Execution time of grayscale conversion